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Education

Polytechnic University of the Philippines, Manila, Philippines: Bachelor of Science in Computer Science
2022 - Present

- Awarded Consistent President's Lister

Higher School of the University of Makati: Science, Technology, Engineering, and Mathematics (STEM) Strand
2020 - 2022

- Graduated with High Honors
- Awarded Overall Champion in Capstone Project

Professional Experience

Legal Assistant, Melba Cawit Law Firm: 2022-Current

- Developed administrative and cross-cultural competencies through exposure to Western legal processes and business correspondence.
- Supported documentation, scheduling, and client coordination, ensuring procedural efficiency and confidentiality in a professional legal setting.

Founding Core Member and Cloud Practitioner, Amazon Web Services (AWS) Cloud Club:
2022-2025

- Initiated the establishment of the AWS Cloud Club to promote cloud literacy and developer collaboration across the PUP network.
- Facilitated technical bootcamps and mentoring sessions for peers.

Full-Stack Developer, Freelance: 2022-Current

- Automated workflows and reviewed product before shipping
- Delivered end-to-end web application solutions integrating front-end design, back-end logic, and database architecture.
- Specialized in Python, JavaScript, Laravel, React, and SQL, creating responsive interfaces and optimized API workflows.

- Implemented modular architecture and scalable deployment strategies aligned with industry best practices.
- Collaborated with clients across diverse sectors, translating functional requirements into high-performance digital systems that emphasized usability, maintainability, and security.
- Consistently refined code quality through version control (Git) and Agile-based iteration cycles, ensuring robust documentation and seamless handovers across projects.

Programming Lead, CISCO Net Connect PUP: 2024-2025

- Directed a multidisciplinary programming team specializing in batch scripting, Python, and JavaScript, overseeing workflow automation and backend optimization initiatives
- Designed modular coding exercises and Git-based collaboration practices to train members in professional version control.
- Provided mentorship on algorithmic thinking and project scalability while ensuring code integrity and maintainability across deployments.

DevOps Intern, Canadian Technology Company Incorporated (CGI): 2025

- Supported CI/CD operations by assisting in build, test, and deployment workflows, ensuring reliable releases across development environments.
- Worked with Git-based version control to manage branches, review changes, and maintain clean, traceable commit history.
- Assisted in environment setup and troubleshooting for Linux-based systems, helping resolve configuration and dependency issues.
- Helped document runbooks and operational procedures to improve handoffs, reduce repetitive manual work, and strengthen team readiness.
- Built the Balik-Bayan Scientist Program System using specialized frameworks and Laravel

Intern and Scholar, Department of Science and Technology – SEI (DOST-SEI): 2025

- Recognized as a national government scholar for exemplary academic performance and innovation potential in computer science.
- Engaged in capacity-building programs promoting scientific research, digital transformation, and public service technology.
- Actively contributed to seminars and workshops advancing AI literacy, open data practices, and ethical computing aligned with DOST's national development goals.
- Supported CI/CD operations by assisting in build, test, and deployment workflows, ensuring reliable releases across development environments.
- Worked with Git-based version control to manage branches, review changes, and maintain clean, traceable commit history.

- Assisted in environment setup and troubleshooting for Linux-based systems, helping resolve configuration and dependency issues.
- Helped document runbooks and operational procedures to improve handoffs, reduce repetitive manual work, and strengthen team readiness.
- Built the Balik-Bayan Scientist Program System using specialized frameworks and Laravel

Technical Adviser and Project Manager, ENC-University of Makati: 2025

- Acted as the lead developer for the project Every Nation BGC One Shared Services system, that also served as the thesis study for an undergraduate group from University of Makati

Technical Care Expert, Sutherland Global Philippines BGC: 2022-2023

- Delivered comprehensive technical support specializing in network, software, and hardware troubleshooting.
- Achieved top performance metrics among peers by demonstrating analytical problem-solving and adaptive learning in a high-pressure environment.
- Enhanced communication and client-handling expertise, fostering measurable customer satisfaction and retention.

Certifications & Professional Training

- Advanced Programming: C, Java, Web Development, PHP, SQL, Python
- Advanced Database Management
- Front-End Development Bootcamp: CSS, Bootstrap, jQuery, JavaScript, React
- Data Science & Machine Learning Foundations
- Nexus Technologies Cybersecurity Training
- AWS Cloud Services and Programming Training
- Cisco Networking & Programming Training

Awards, Recognition & Academic Distinctions

- Overall Best Capstone Project, University of Makati – Senior High School
- Capstone Project Champion (2022)
- President's Lister, Polytechnic University of the Philippines (2022–2025)
- Graduated with High Honors, Senior High School (2022)
- Most Excellent Trainee, Sutherland BGC (2022)
- Dean's Lister & President's Lister, Fort Bonifacio High School (2020)
- Founding Member, AWS Cloud Club – PUP (2023)
- Programming Lead, Cisco NetConnect Program (2024)

Scholarship Grants & Competitive Academic Selection

- Department of Science and Technology-SEI Undergraduate Scholarship
- Real LIFE Foundation Private Scholarship
- LANI Local Government Scholarship
- NEON Foundation Private Scholarship
- Qualified for Admission, Rizal Technological University (BS in Computer Science)
- Qualified for Admission, University of Makati (BS in Computer Science)
- Qualified for Admission, Pamantasan ng Lungsod ng Manila (BS in Computer Science)
- Regional Journalism in English Sportswriting Qualifier

Technical Skills

Programming Languages

- Python, JavaScript, TypeScript, Java, C, C++, C#, SQL, PHP, Bash

Web Technologies

- HTML5, CSS3

Frameworks & Libraries

- React, Vue.js, Node.js, Express.js, Laravel, Bootstrap, Django, Flask
- Pandas, NumPy, PyTorch, TensorFlow, scikit-learn

Databases

- MySQL, PostgreSQL, SQLite, Firebase, MongoDB

Cloud & Infrastructure

- Amazon Web Services (AWS): EC2, S3, RDS, IAM, Lambda (basic)

DevOps & Version Control

- Git, GitHub, Docker, CI/CD pipelines

APIs & Integration

- RESTful API design, GraphQL (basic), JSON

Automation & Tooling

- Shell scripting, command-line tools, build automation

Front-End Practices

- Responsive design, cross-browser compatibility

Software Engineering & Development Leadership

- Led Agile/Scrum workflows including sprint planning, task breakdown, and delivery coordination
- Drove system design decisions and modular architecture planning for scalable applications
- Acted as a technical liaison with stakeholders to align requirements, scope, and timelines
- Partnered with UI/UX designers to improve usability and overall user experience

Leadership & Professional Skills

- Team leadership and mentoring of engineers, including onboarding and skill development
- Facilitated cross-functional collaboration and resolved technical or delivery conflicts
- Execution-focused planning with accountability for outcomes and deadlines
- Client communication, technical presentations, and progress reporting

Projects

IntelliForm – Intelligent PDF Form Understanding System (2024–2025)

IntelliForm is an AI-driven document intelligence system designed to understand and extract structured information from complex government, banking, and healthcare PDF forms. The system combines LayoutLMv3 for layout-aware document understanding, Graph Neural Networks to model structured relationships between form fields, and a T5-based summarization component for both field-level and document-level abstraction. The solution was architected modularly, separating field classification, graph construction, and summarization pipelines, and delivered production-style inference using FastAPI alongside a lightweight static JavaScript interface for assisted form completion and review. Evaluation followed research-grade standards using F1 score, Intersection over Union (IoU), ROUGE, and Expected Calibration Error (ECE), supported by a monitoring dashboard to track accuracy, calibration behavior, and field-level performance for iterative improvement.

Balik Scientist Program Management System (BSPMS) – DOST (2025)

This project involved contributing to a Laravel-based government information system supporting long-term reporting for the Balik Scientist Program. The work focused on report automation, engagement analytics, and PDF report generation for executive-level outputs. Key contributions included implementing Report #12 (List of BSP Awardees) and Report #16 (Summary Report of BSP) using optimized raw SQL with complex multi-table joins across engagements, host institutions, and scientific councils. The system produced standardized, executive-ready report templates with aggregated rollups and merged headers, enabling reliable statistical analysis across a ten-year reporting horizon.

Dataset Construction Pipeline for Philippine Government Forms (2025)

A reproducible data-processing pipeline was engineered in Python to convert annotated PDF forms into structured JSON datasets suitable for training and evaluating LayoutLMv3 models. The pipeline automated OCR alignment, bounding-box extraction, BIO tagging, and hash-based canonical ID generation to ensure consistent labeling, stable identifiers across revisions, and reliable dataset traceability. This tooling supported scalable experimentation, dataset versioning, and research-grade reproducibility for document AI workflows.

ONE Shared Service – Every Nation BGC (2025)

This project involved leading the development of a private, web-based one-stop booking system for Every Nation Bonifacio Global City. The platform was fully deployed to support internal requests for manpower allocation, shuttle services, and room reservations. The system emphasized usability, operational efficiency, and centralized scheduling to streamline organizational workflows.

Barangay San Miguel Queue Management System (2025)

This public-facing web application was developed to modernize community document processing for Barangay San Miguel. The system integrated an IoT-enabled kiosk device to automate queue handling and residential workflows, reducing manual processing and improving service efficiency. The project focused on real-world deployment considerations, including reliability, accessibility, and workflow automation for local government use.

CISCO NetConnect Program – Programming Lead (2024–2025)

As Programming Lead, this role involved directing a team of student developers focused on building practical engineering fundamentals through batch scripting, Python, and JavaScript. Responsibilities included standardizing Git-based collaboration workflows, leading structured code review sessions, and reinforcing best practices in version control, automation, and algorithmic problem-solving. The program emphasized disciplined software development habits and collaborative engineering culture.

Freelance Full-Stack Development Projects (2023–Present)

Delivered end-to-end web solutions for small businesses and early-stage teams, working across Laravel, React, Node.js/Express, and SQL-backed systems. Projects included secure authentication, maintainable REST APIs, responsive front-end delivery using Bootstrap, and deployment pipeline implementation. Each engagement was supported by clear technical documentation and stakeholder communication to ensure reliable delivery and measurable outcomes.

AWS Cloud Club PUP Initiatives (2022–2025)

Founded and co-developed the digital infrastructure for AWS Cloud Club – Polytechnic University of the Philippines. Contributions included building the organization’s official website, an interactive photobooth web application, and internal event resource management tools. Systems were deployed cloud-first using Next.js and AWS services such as S3, CloudFront, and Lambda, with CI/CD automation implemented through GitHub Actions to ensure secure, repeatable deployments for student-led events and workshops.

Utility and Automation Tools Suite (2022–2025)

Developed automation-centric operational systems using Zapier to orchestrate end-to-end workflows involving intake, processing, notifications, and reporting. Implemented multi-step Zaps utilizing Webhooks, Paths, Filters, Formatter, Schedule, and Storage, alongside production-minded safeguards such as deduplication keys, idempotent execution patterns, structured error handling, retries, and fallbacks. Extended workflows with custom code steps for validation and transformation, and delivered reusable automation templates and runbooks emphasizing maintainability, least-privilege access, and secrets hygiene.

Capstone Project – BioLabs (2022)

Led a capstone team recognized as Overall Best Project Champion for developing an augmented reality-enabled prototype that simulates science laboratory classes. The project focused on immersive instructional design and interactive workflows to support experiential learning in environments with limited physical laboratory resources.

4PH Pag-IBIG Housing Loan Management System (2025)

Designed and developed a housing loan management system inspired by Pag-IBIG Fund workflows, emphasizing secure and efficient handling of classified user and loan information. The system applied robust database administration practices, optimized schema design, and efficient data retrieval strategies to ensure reliable tracking of user profiles and loan records.

Ricochet Rival (2025)

Ricochet Rival is a 2D shooting game featuring artificial intelligence driven by Monte Carlo-based decision strategies. The game supports both player-versus-player and player-versus-AI modes

with adjustable difficulty levels. The AI agent learns from environmental states and player behavior, dynamically adapting over time to deliver progressively challenging gameplay.

Letter Website Series (2025)

This series consists of personalized letter-album websites created to express gratitude and meaningful messages to loved ones. Each site applies modern front-end design principles, emphasizing emotional storytelling, responsive layouts, and visual consistency to deliver a memorable and intimate user experience.

Moon Exhibit Website (2025)

An educational web project designed to teach users about the phases of the moon through creative and interactive presentation. The website applies modern design techniques and visual storytelling to make astronomical concepts accessible, engaging, and easy to understand.

Quiz Website – Pokémon Edition (2025)

A Game Boy–inspired Pokémon quiz game developed using modern web technologies and the Laravel framework. The application manages quiz logic and user progress through session storage, blending classic gaming aesthetics with contemporary usability and performance considerations.

Manam Restaurant Food Menu Website (2025)

Developed a full-featured restaurant website using Laravel to creatively showcase Manam’s food offerings. The platform emphasizes organized menu structure, strong visual presentation, and responsive design to deliver an engaging digital dining experience.

Kael’s Adventure Land Catalogue System (2025)

Built a fully functional web-based catalog system for an inflatable entertainment business using Laravel. The platform includes both customer-facing and administrative interfaces, enabling efficient content management, product listings, and customer inquiries through a clean and interactive interface.

Prismatic (2024)

Applied the principles of programming language to develop a new programming language based on the structure of the programming language C. It is capable of lexical and semantic programming

Merc-Airline Ticketing System (2024)

A Java-based desktop application simulating a modern airline ticketing system using a graphical user interface. The project demonstrates core object-oriented programming principles while modeling real-world booking workflows, including ticket selection, pricing logic, and passenger information handling.

Trashketball (2024)

A 3D basketball shooting game developed in C#, focusing on physics-based interactions, player controls, and core gameplay mechanics within a three-dimensional environment.

Unbroke (2024)

A Java application designed to support personal budget management by allowing users to track expenses and improve financial habits. The project emphasizes usability, logical flow, and practical problem-solving.

SmartPlate (2024)

An application that recommends locally available food recipes tailored for dormitory residents around PUP-Manila. The project focuses on accessibility, affordability, and practicality to support students in daily meal planning.

Real LIFE Foundation Scholarship Application Website

Contributed to the development of the Real LIFE Foundation's scholarship application platform using Laravel. The system includes applicant and administrative interfaces, supporting registration, application submission, review workflows, and streamlined scholarship management.